Mohamed Amine Darraj

AI / Data — Final-Year Internship

6 months · Available late March / early April 2027

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PROFILE

Final-year engineering student pursuing a double degree at École Centrale de Lyon, specializing in artificial intelligence and data engineering. Passionate about designing intelligent solutions, from data exploration to the deployment of complete applications. Hands-on experience in machine learning, generative AI and full-stack development, with an approach combining technical rigor, creativity and real-world problem solving.

EDUCATION

École Centrale de Lyon *2026–2027*
General engineering double-degree program; M2 Computer Science — Artificial Intelligence.

National Higher School of Arts and Crafts, Meknes *2024–2027*
Engineering cycle — Artificial Intelligence and Data Technologies.

National Higher School of Arts and Crafts, Meknes · Integrated preparatory cycle *2022–2024*

PROFESSIONAL EXPERIENCE

AI Engineer Intern · NOMAC — Noor III Solar Power Plant *Summer 2026*

- Built **Noor Alerts**, an industrial AI platform using a corpus of **28,692 SCADA and maintenance events**, with FastAPI, React and PostgreSQL.
- Achieved **95.2% hold-out classification accuracy** with TF-IDF and Logistic Regression; integrated a **GraphRAG** copilot combining Neo4j, FAISS and an LLM for incident retrieval.

IT Intern · YAZAKI Meknès *Summer 2025*

- Built an **SNMP-based printer monitoring dashboard** automating equipment-status collection and incident detection.

PROJECTS

ProAct CMMS — AI-Augmented Maintenance · *FastAPI, Next.js, PostgreSQL, FAISS* github.com/AM1N8/GMAO-Project

- Built a full-stack CMMS (equipment, work orders, inventory, technicians, FMECA, maintenance KPIs) with an LLM copilot (**RAG over technical manuals**), semantic retrieval, OCR-assisted extraction and predictive maintenance.

MUSIQ — Reinforcement Learning for Music Recommendation · *PyTorch, Gymnasium* github.com/AM1N8/MUSIQ-

- Compared **DQN, A2C, SAC and CQL** in a Gymnasium-compatible environment using discrete song selection and continuous embedding-based action spaces; designed Dueling DQN, attention and Transformer agents with multi-objective rewards balancing engagement, diversity and personalization.

Computer Vision for Healthcare · *YOLOv8, ResNet50, NLP* github.com/AM1N8/Computer-Vision-HealthCare

- Built chest X-ray abnormality-localization pipelines using **YOLOv8** and bone-fracture classifiers using **ResNet50/CNN**; integrated a symptom-analysis assistant using TF-IDF, KNN and cosine similarity.

Multimodal Trajectory Prediction · *TensorFlow, LiDAR, BiLSTM* github.com/AM1N8/TrajectoryPredictionProject

- Combined a **PointNet-inspired LiDAR encoder** with a **BiLSTM and self-attention**; engineered motion features (speed, acceleration, jerk, heading, curvature) and built a Streamlit inference and visualization interface.

TECHNICAL SKILLS

Programming: Python, SQL, C/C++, R, Java, Bash
Machine Learning / Deep Learning: PyTorch, TensorFlow, scikit-learn, Hugging Face, OpenCV, SHAP, LIME
LLMs / Data / MLOps: LangChain, LlamaIndex, FAISS, Neo4j, Pandas, NumPy, Polars, MLflow, Optuna
Engineering: FastAPI, PostgreSQL, Redis, Docker, Git, Linux, React, Next.js, Streamlit

AWARDS AND ACTIVITIES

HackAI 4.0 — 5th Place · 1337AI *2026*

- Built **Hirfati**, a multilingual Darija/Tamazight pipeline combining speech recognition, structured-data extraction, RAG-based price estimation and MCP publishing workflows.

Gadz'IT — Development Cell *2022–Present*

- Built **DataTUI**, a Polars-based terminal EDA toolkit, and 2D/3D game and VR prototypes with Godot and Unity.

LANGUAGES & PASSIONS

Languages: French & English (Professional) · Arabic & Tamazight (Native)
Passions: Game development & gaming culture · Music, sound design & audio exploration · Cinema & immersive storytelling